

Problem 11.1 What exoplanets do we know?

Explore the online archive exoplanets.org and/or exoplanetarchive.ipac.caltech.edu to get an overview of what exoplanets have been found so far. Make different plots to get an overview of the distribution of exoplanets, looking at their masses, radii, orbital distances, and any other parameter you find interesting. Discuss the trends that you discover. Which could be caused by planet formation and migration mechanisms? Which might be due to the use of different detection methods and their inherent bias?

Hint: You can use the online plotting tools provided by the websites (faster way) or access the chosen database using **astroquery** (preferred if you also want to gain some experience with this very useful package).

Problem 11.2 Relative velocity

A planet embryo and a planetesimal orbit a star at distances a and $a + \Delta a$. Show that the relative velocity between these two objects (as an absolute value) can be expressed as

$$v_{rel} \sim \Omega \Delta a. \quad (4)$$

Hint: Start by expressing their velocities using the Keplerian velocity Ω at their specific distance. Use $\Delta a/a \ll 1$ in your calculation.